5th

(*) Why transition elements have high (*) hydration energy?

(*) The transition metal ions are small in size and high charge Due to this size/charge ratio (charge density) is high, hence dgn hydration energy also high.

(*) In hydrated ion of a transition element not only an ion dipole bond is possible but also there is tendency toward co-ordinate bond formation to fill incomplete orbitals of the cent ion. Since process may be involved in the formation of a hydrated transition ion it would seem probable that such an ion should be more stable than a hydrated ion in which only ion-dipole forces are involved.

Thus, hydrated transition elements have enthalpy of hydration.

(*) Why $Cu^{2+}_{(aq)}$ is more stable than $Cu^{+}_{(aq)}$

→ Stability depends on the hydration energy (enthalpy) of the ions when they bond to the water molecules. The Cu^{2+} ion has a greater charge density than Cu^{+} ion and thus

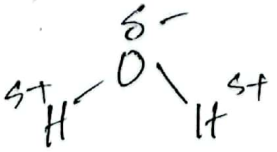
form much stronger bonds releasing more energy. The extra energy needed for the second ionization of the + copper is more than compensated for by the hydration, so much so that the Cu^+ ion loses an electron to become Cu^{2+} which can then release this hydration energy —

In aqueous solution Cu^+ disproportionates to Cu^{2+} and Cu

$$2Cu^+ \longrightarrow Cu^{2+} + Cu^+$$

(*) Lattice energy, Solvation, Hydration energy.
↓
Energy released when one mole of ions (in gaseous phase) are brought together from infinity.

(*) Energy released when one mole of ions undergo hydration (in water H_2O).



Hydrogen bonding, dipole-dipole, ion-induced dipole - dipole induced dipole.

$$\Delta H_{hyd} = \Delta H_{lattice} + \Delta H_{sol}$$
$$-774 \quad 778 \quad +4$$

(*) The molar enthalpy of solution, ΔH_{sol}, is the energy released when one mole of solid is dissolved in a solvent.

⊕ endothermic
⊖ exothermic.

$Na^+(s) \longrightarrow Na^+(g)$ $\quad \Delta H^{\ominus}_{Hyd} = -406\ kJ$.

$Cl^-(s) \longrightarrow Cl^-(aq)$ $\quad \Delta H^{\ominus}_{Hyd} = -377\ kJ$.

Tutorial on Ions

Co-ordination chemistry (11)

(*) The stability of Co-ordination Compounds

(*) The formation of complexes in aqueous soln is a matter of great importance not only in inorganic chemistry but also in biochemistry, analytical chemistry and in a variety of application.

Suppose we put a metal ion M and some monodentate ligand L, together in soln. If we assume that no insoluble products or any species containing more than one metal ion are formed, then equilibrium expressions of the following sort will describe the system.

$$M + L = ML \qquad K_1 = \frac{[ML]}{[M][L]}$$

$$ML + L = ML_2 \qquad K_2 = \frac{[ML_2]}{[ML][L]}$$

$$ML_2 + L = ML_3 \qquad K_3 = \frac{[ML_3]}{[ML_2][L]}$$

$$\vdots \qquad \vdots$$

$$ML_{N-1} + L = ML_N \qquad K_N = \frac{[ML_N]}{[ML_{N-1}][L]}$$

Basic inorganic chemistry (third ed)
F. Albert cotton, G. Wilkinson, P.L. Gaus.

There will be N such equilibria, where N represents the maximum co-ordination number of the metal ion M for the ligand L. The parameter N may vary from one ligand to another. For example

Al^{3+} forms $AlCl_4^-$ and AlF_6^{3-}

Co^{2+} " $CoCl_4^{2-}$ and $Co(NH_3)_6^{2+}$

as the highest complexes with the ligand indicated.

Another way of expressing the equilibria relations is the following.

$$M + L = ML \qquad \beta_1 = \frac{[ML]}{[M][L]}$$

$$M + 2L = ML_2 \qquad \beta_2 = \frac{[ML_2]}{[M][L]^2}$$

$$M + 3L = ML_3 \qquad \beta_3 = \frac{[ML_3]}{[M][L]^3}$$

$$\vdots$$

$$M + NL = ML_N \qquad \beta_N = \frac{[ML_N]}{[M][L]^N}$$

Since there can be only N independent equilibria in such a system, it is clear that the K_i's and the β_i's must be readily related.

Let us consider for example the expression for β_3. Let us multiply both numerator and denominator by $[ML][ML_2]$ and then rearrange slightly.

$$\beta_3 = \frac{[ML_3]}{[M][L]^3} \cdot \frac{[M][L][ML_2]}{[ML][ML_2]}$$

$$= \frac{[ML]}{[M][L]} \cdot \frac{[ML_2]}{[ML][L]} \cdot \frac{[ML_3]}{[ML_2][L]}$$

$$= K_1 \cdot K_2 \cdot K_3 .$$

It is not difficult to see that this kind of relationship is perfectly general namely.

$$\beta_K = K_1 K_2 K_3 \ldots K_K = \prod_{i=1}^{i=K} K_i$$

The K_i's are called the stepwise formation constants (or stepwise stability constants) and β_i's are called the overall formation constants (or overall stability constants)

$$Cd^{2+} + NH_3 = [Cd(NH_3)]^{2+} \quad K = 10^{2.07}$$

$$[Cd(NH_3)]^{2+} + NH_3 = [Cd(NH_3)_2]^{2+} \quad K = 10^{2.10}$$

$$[Cd(NH_3)_2]^{2+} + NH_3 = [Cd(NH_3)_3]^{2+} \quad K = 10^{1.44}$$

$$[Cd(NH_3)_3]^{2+} + NH_3 = [Cd(NH_3)_4]^{2+} \quad K = 10^{0.93}$$

$$\beta_4 = 10^{7.12}$$

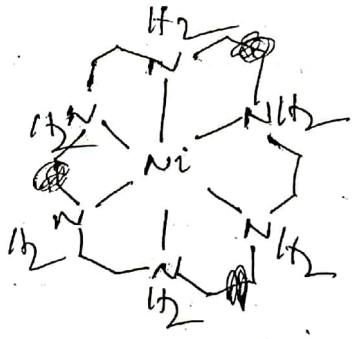
Thus, as ligand is added to the soln of metal ion, ML first forms more predominantly than any other complex in the series. As addition of ligands is continued, the ML_2 concentration rises rapidly, while the ML conc drops. Then ML_3 becomes dominant, ML and ML_2 become unimportant and so on, until the highest complex ML_N is formed to the nearly complete exclusion of all others at very high ligand conc.

The chelate effect —

As a general rule, a complex containing one (or more) five or six membered chelate rings is more stable (has a higher β formation constant)

$$Ni^{2+}(aq) + 6\,NH_3(aq) = [Ni(NH_3)_6]^{2+}(aq) \qquad \beta_6 = 10^{8.6}$$

$$Ni^{2+}(aq) + 3\,H_2NCH_2CH_2NH_2(aq) =$$



$$\beta_3 = 10^{18.3}$$

→ than a complex that is as similar as possible, but lacks some or all of the chelate rings.

The complex with three 5 chelate rings is about 10^{10} times more stable.

⊗ Thermodynamic stability —

$$Ni(NH_3)_6^{2+}(aq) + 3(en)(aq) = Ni(en)_3^{2+}(aq) + 6NH_3(aq)$$

en = ethylenediamine.

$$K = 10^{9.7}$$

$$\Delta G^\circ = -RT\ln K = -67 KJ mol^{-1} = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta H^\circ = -12 KJ mol^{-1}$$

$$-T\Delta S^\circ = -55 KJ mol^{-1}$$

It is evident that both enthalpy and entropy favor the chelate complex, but the entropy contribution is far more important The chelate effect is essentially an entropy effect.

→ The nickel ion is co-ordinated by six H_2O molecules. In each of the first two reactions, These six H_2O molecules are liberated when nitrogen ligands become co-ordinated. On that point of view two processes are equivalent.

However, in one case six NH_3 molecules lose their freedom at the same time, and there is no net change in the number of particles. In the other case, only three en molecules lose their freedom, and thus there is a net increase of 3 mol of individual molecules. The reaction with three en causes a much greater increase in disorder than does that with six NH_3 molecules and therefore ΔS° is more positive (more favorable) in the former case than in the latter.

→ Another way to state the matter is ↑ ⟷ the chelate effect weakens as ring size increases. The effect is greatest at five and six-membered rings, becomes marginal for seven-membered rings and is unimportant thereafter.

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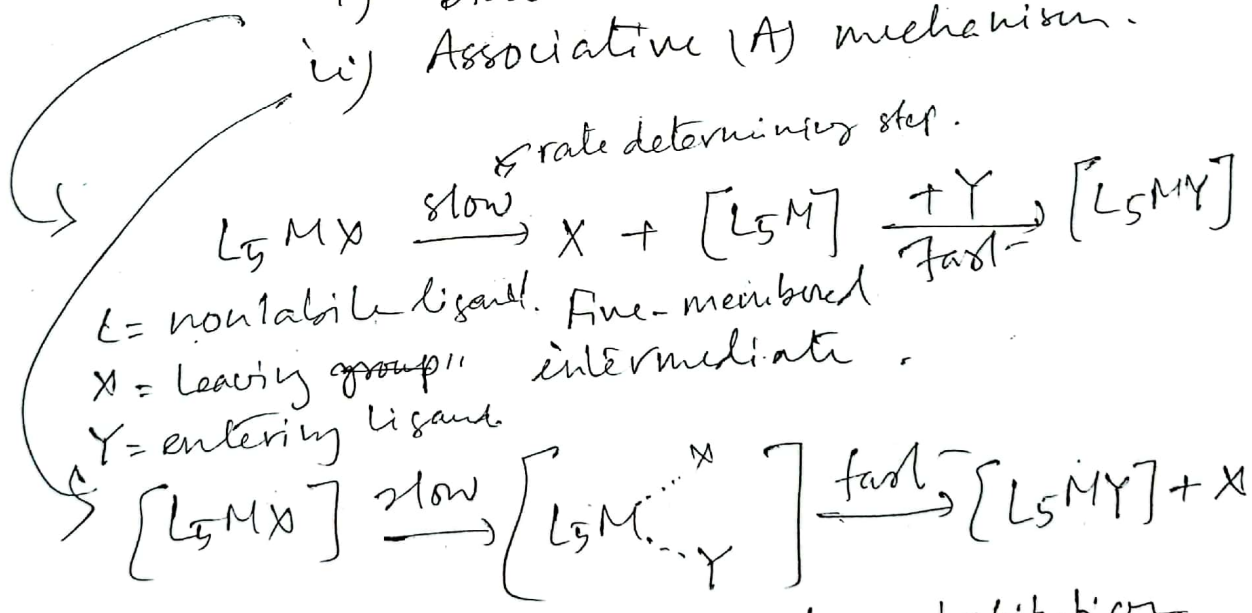
Reactivity of Co-ordination Compounds. -

Substitution Reactions of Octahedral Complexes.

Keys words - labile ; inert ; stable ; unstable

→ Two extreme mechanistic possibilities may be considered for any ligand substitution. or any single step in a series of substitution reaction.

i) Dissociative (D) mechanism.
ii) Associative (A) mechanism.



$$L_5MX \xrightarrow{\text{slow}} X + [L_5M] \xrightarrow[\text{fast}]{+Y} [L_5MY]$$

$$[L_5MX] \xrightarrow{\text{slow}} \left[L_5M\begin{matrix}\cdots X\\ \cdots Y\end{matrix}\right] \xrightarrow{\text{fast}} [L_5MY] + X$$

(i) This mechanism for ligand substitution is comparable to the S_N1 mechanism in

organic systems, because the formation of the intermediate with reduced co-ordination number is unimolecular, as well as rate determining.

The interchange of the ligands X and Y could be accomplished mostly by breaking the bond to the leaving group.
→ Id (interchange-dissociative)
→ I_A (interchange-associative)
or by making the bond to the entering group.

⊛ To complicate matters further, the rate law that is determined for a reaction from kinetic data cannot be used to identify the mechanism for that reaction. This is so because additional steps in the overall substitution may take place, obscuring the simple first- and second-order rate laws that are expected for unimolecular and bimolecular processes, respectively.

The three most important cases that illustrate this sort of complication are

i) Solvent intervention.

ii) Ion-pair formation.

iii) Conjugate-base formation.

i) Solvent intervention —

Conc[H_2O] [~55.5M]

$$[L_5MX] + H_2O \longrightarrow [L_5MH_2O]^{\oplus} + X^{\ominus} \text{ (slow)}$$

$$[L_5MH_2O] + Y \longrightarrow [L_5MY] + H_2O \text{ (fast)}$$

A simple first-order rate law would be observed and yet either reaction could proceed by an A (or Ia) or D (or Id) mechanism. Intervention of the solvent in reaction obscures the molecularity of the rate-determining step; the reaction will necessarily be observed to be first order because of the high and constant concentration of the entering ligand H_2O.

(ii) Ion-pair Formation

when the reacting complex and the entering ligand are both ions, especially when both have high charges, ion pairs (or outersphere complexes, as they are called) will form

$$[L_5MX]^{n+} + Y^{m-} = \{[L_5MX]Y\}^{n-m}$$

In the product the entering ligand Y has been stabilized at the outer edge of the co-ordination sphere of the complex $[L_5MX]^{n+}$ primarily by electrostatics.

(iii) Conjugate-base Formation —

when experimental rate laws contain $[OH^-]$ there is the question whether OH^- actually attacks the metal in a true association fashion, or whether it appears in the rate law through

operation of mechanism shown below –

$$[Co(NH_3)_5Cl]^{2+} + OH^- \longrightarrow [Co(NH_3)_4(NH_2)Cl]^+ + H_2O$$

$$[Co(NH_3)_4(NH_2)Cl]^+ \xrightarrow[then\ H^+]{+Y^-} [Co(NH_3)_5Y]^{2+} + Cl^-$$

In this conjugate-base (CB) mechanism, the ~~OH~~ hydroxide first ~~depot~~ deprotonates a ligand (usually NH_3) forming the conjugate base, here leading to the NH_2^- ligand. It is then the conjugate base of the original metal complex that reacts with the incoming ligand.

Substitution Reactions in Square Complex.

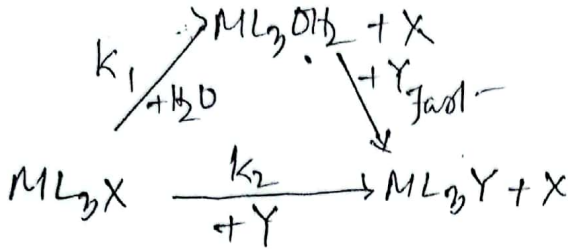
For square complexes, the mechanistic problem is more straightforward and best better understood.

One might expect that four-coordinate complexes would be more likely than octahedral complexes to react by associative pathways because the co-ordination sphere is less congested to start with. Extensive studies of Pt^{II} complexes have shown that this is true.

$$PtL_3X + Y \longrightarrow PtL_3Y + X$$

L represents nonlabile ligands, X is the leaving ligand and Y is the entering ligand. Reactions of the type shown in reaction are found to have the rate law shown.

$$rate = k_1[PtL_3X] + k_2[PtL_3X][Y]$$



The R.D. k_1 path involves rate-determining formation of the aquated intermediate $[PtL_3(OH)_2]^{n+}$ and subsequent, rapid substitution of the aqua ligand by Y. Both of these steps in the k_1 path appear to involve associative activation.

The k_2 path involves direct replacement of X by Y also via associative activation.

(*) Charge effect
Steric "
Entering Ligand.

Stereochemistry

(X)

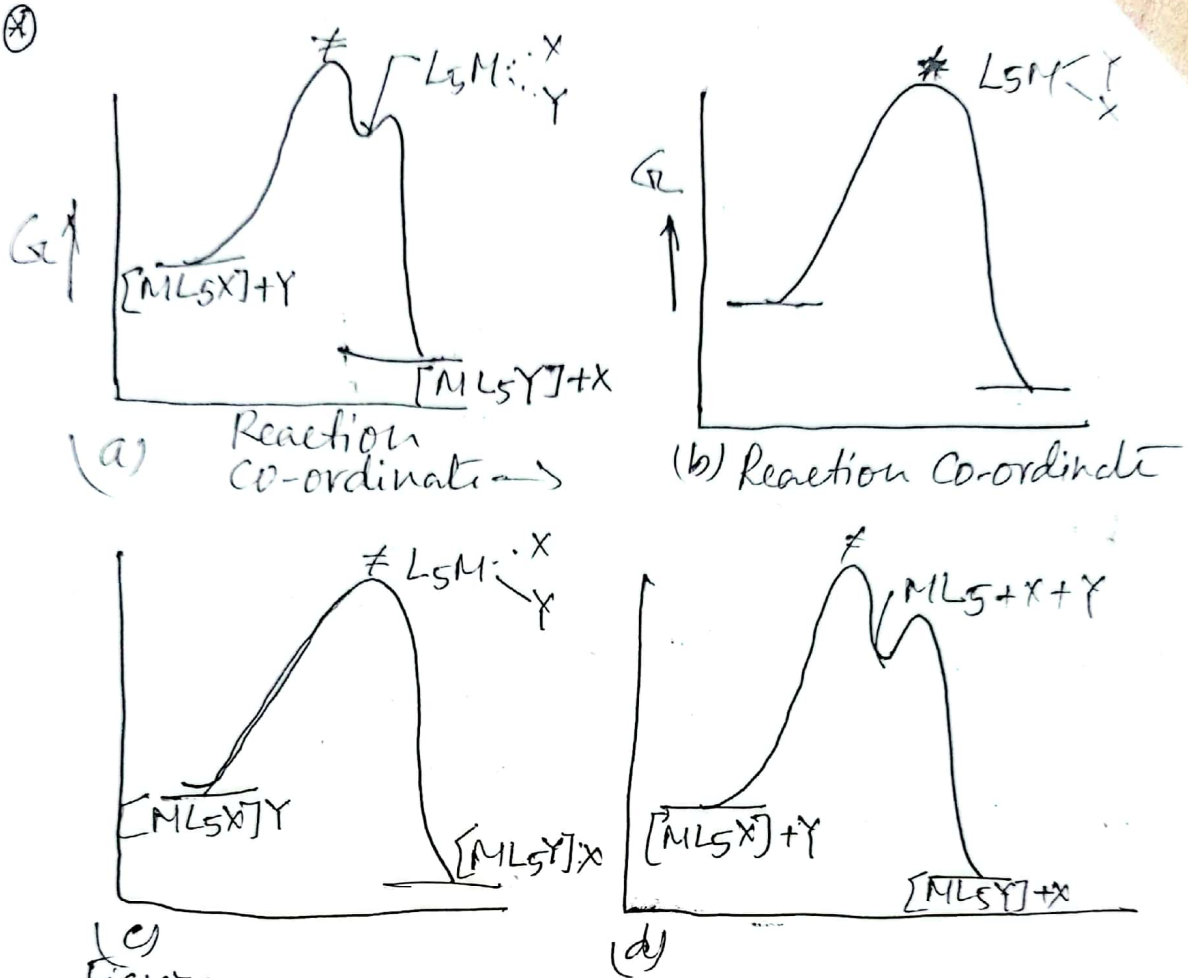


Figure - The four general mechanisms for ligand substitution in the complexes $[ML_5X]$, where L are nonlabile ligands, X is the leaving ligand and Y is the entering ligand.

(a) The associative (A) mechanism in which an intermediate of expanded co-ordination number is formed first - through rate-determining entry of the ligand Y.

(b) The interchange-associative mechanism (Ia) in which the transition state is reached mostly through formation of the bond (M···Y) to the entering Ligand.

The notations $[ML_5X];Y$ and $[ML_5Y];X$ for the reactants and product, respectively represents outer-sphere complexes (or ion pairs) as formed in reaction 6.5.7.

(c) The interchange-dissociative mechanism (Id). in which the transition state is reached mostly through breaking the bond (M···x) to the leaving Ligand. Again, the reactants and products are outer-sphere complexes (or ion pairs) as featured in Reaction 6.5.7.

(d) The dissociative or D mechanism in which an intermediate of reduced co-ordination number is formed first through rate-determining cleavage of the bond to the leaving group.

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Charge Effects:-

Consider the series of Pt^{II} complexes with charges varying from +1 to −2.

$[Pt(NH_3)_3Cl]^+$ $[Pt(NH_3)_2Cl_2]$

$[Pt(NH_3)Cl_3]^-$ $[PtCl_4]^{2-}$

→ The observed rate constants k_1 (for aquation in water solvent) vary only by a factor of two. This is a remarkably small variation, given the large differences in charge among the complexes.

Pt–Cl bond breaking — difficult in the higher positive charge

Also, complexes with the higher positive charge should favor approach of the nucleophile. Since neither of these trends is observed, an associative process is indicated in which both Pt–Cl bond breaking and Pt–OH_2 bond making are of comparable

importance.

Steric effects – Steric acceleration is observed for substitution reactions of octahedral complexes, and this was taken to be evidence for a dissociative nature in such reactions. For square complexes, substitution reaction are retarded by steric crowding at the metal ~~centr~~ center. This is taken to be evidence that the entering ligand Y must approach the metal centre in order to reach transition state. This result is consistent with either an associative (A) or Ia interchange

Entering Ligand Effects. The second-order rate constant k_2 in the equation is strongly dependent on the nature of the enter ligand. A reactivity series can be established in which the entering ligands Y are placed in order depending on the value of k_2.

$F^- \sim H_2O \sim OH^- < Cl^- < Br^- \sim NH_3 \sim$ alkenes $<$ $C_6H_5NH_2 < C_5H_5N < NO_2^- < N_3^- < I^- \sim SCN^- \sim R_3P$.

This is essentially the order of nucleophilicity towards Pt II that is expected for these ligand and an associative mechanism ~~in~~ is indicated.

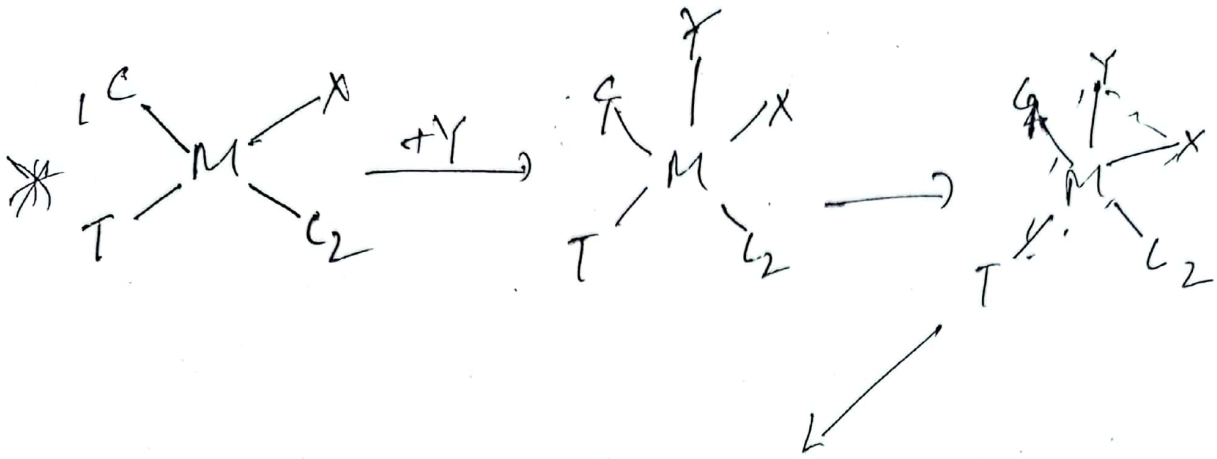
Nonlabile Ligands — The trans effect.

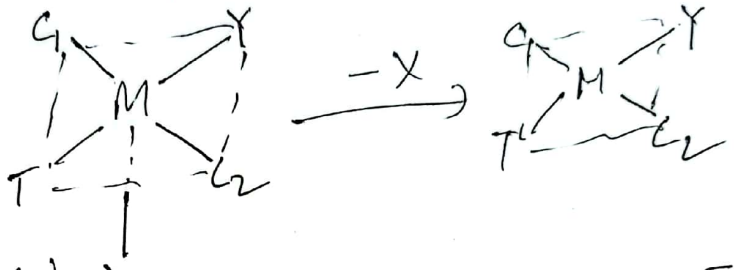
$$[PtLX_3] + Y \longrightarrow [PtLX_2Y] + X$$

A particular feature of substitution at square complexes is the important role played by nonlabile ligands that are trans to the leaving ligand.

$$[PtLX_3] + Y = [PtLX_2Y] + X$$

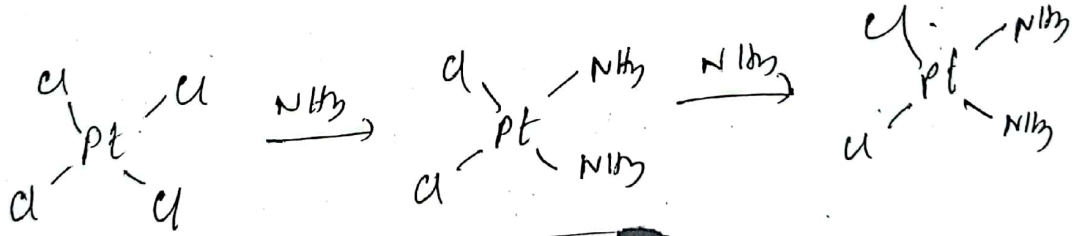
Any one of three labile ligands X can be replaced by the entering ligand Y. Furthermore, the ligand X that is replaced can be either cis or trans to L, leading to cis-trans orientation of Y with respect to L in the product. It has been found that the relative proportions of cis and trans products varies appreciably with the nature of the ligand L.

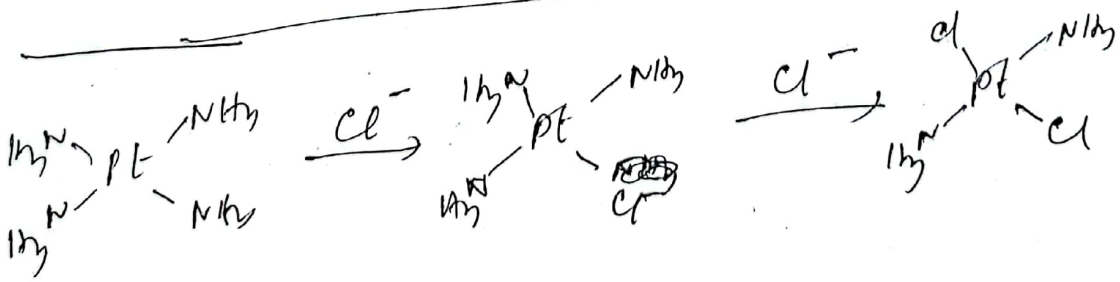


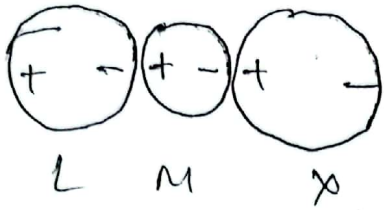


(explanation next page) X

H_2O, OH^-, NH_3, py (NC5H5) < Cl^-, Br^- < SCN^-, I^-, NO_2^-, $C_6H_5^-$ < $SC(NH_2)_2$, CH_3^- < H^-, PR_3 < C_2H_4, CN^-, CO ..







The arrangement of dipoles along the trans L–M–X axis according to the polarization theory of the trans effect.

---

* The se steric course of ligand substitution in square complexes and the structure (trigonal bipyramidal) of the five-co-ordinated intermediate. The ligand C_1 or C_2 occupy axial positions in the intermediate trigonal bipyramid are the ligands that are cis to the leaving group X in the reactant. The ligand T in the reactant is the strongest trans director as it lies trans to X the leaving ligand X. The entering ligand Y, the leaving X, and the trans ligand T, share the equatorial positions of the trigonal

---

bipyramidal intermediate. The new ligand Y in the product occupies the co-ordination position that was vacated by the leaving ligand X.

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bipyramidal intermediate. The new ligand r in the product occupies the co-ordination position that was vacated by the leaving ligand X.

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Electron-transfer reaction.

→ Oxidation-reduction (redox) reactions in which an electron passes from one complex to another.

Example.

→ $Fe^{2+}_{(aq)} + Ce^{4+}_{(aq)} \longrightarrow Fe^{3+}_{(aq)} + Ce^{3+}_{(aq)}$

in which the aqua ion of Ce^{IV} is reduced by the aqua ion of Fe^{II}

→ An electron-transfer reaction may take place so that there is actually no net chemical change, as in

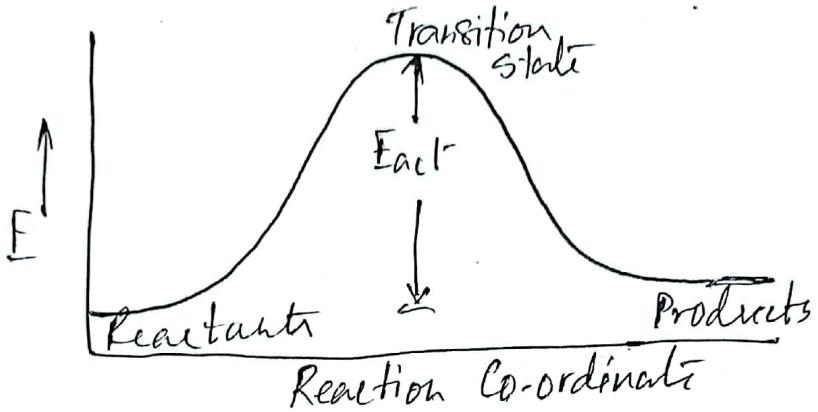
$$[*Fe(CN)_6]^{4-} + [Fe(CN)_6]^{3-} = [*Fe(CN)_6]^{3-} + [Fe(CN)_6]^{4-}$$

→ Reaction such as are called self-exchange reactions →

→ Self-exchange reactions can only be followed by using isotopic tracers or certain magnetic resonance techniques.

→ These reactions are of interest because there is no change in free

energy as consequence of reaction and the free energy profile is symmetrical.



→ Two well established general mechanisms.

Outer-sphere mechanism → only the outer or solvent, co-ordination sphere of the two metal complexes are displaced during the reaction.

→ No substitution of the ligands in the inner co-ordination spheres of either reactant is needed in order for electron transfer to take place. (There are required changes in metal-ligand bond length, however;)

Inner-sphere electron-transfer → the inner-coordination sphere of one reactant must first undergo substitution to accept a new ligand

→ The new ligand must serve, once substitution has taken place, to bridge the two metal ~~centr~~ centers together. This bridging ligand is bound to the inner-co-ordination spheres of both metal center.

OS mechanism →

$$[Fe^{II}(CN)_6]^{4-} + [Ir^{IV}Cl_6]^{2-} \rightleftharpoons [Fe^{III}(CN)_6]^{3-} + [Ir^{III}Cl_6]^{3-}$$

- Complexes participating in the reaction undergo ligand substition reactions more slowly than they participate in electron transfer reaction.

(where both reactants are inert towards substitution ($t_{1/2} > 1m$)

but the redox reaction is fast ($k = 10^5 L mol^{-1} s^{-1}$)

$$[Fe(CN)_6]^{4-} + [IrCl_6]^{2-} = [Fe(CN)_6]^{4-}/IrCl_6]^{2-}_{K_{os}}$$

$$[Fe(CN)_6]^{4-}/IrCl_6]^{2-} \longrightarrow [Fe(CN)_6]^{3-} + [IrCl_6]^{3-}_{ket.}$$

The two steps of a general outer-sphere mechanism can be illustrated using.

-step 1 - There is a preequilibrium, characterized by the constant K_{os}, in which an outer-sphere complex (or ion pair) is formed.

Step 2 -
→ This encounter (outer-sphere complex or ion pair) between the reactants brings them to within the internuclear separation required for electron transfer.

→ ~~This~~ The electron-transfer step takes place within this outer-sphere complex, only after metal-ligand bond lengths have been altered enough to allow the electron transfer to take place adiabatically (without further change in energy).

The Inner-Sphere (or Ligand-bridged) Mech

Ligand-bridged transition states have been shown to occur in a number of reactions, mainly through the elegant experiments devised by H. Taube and his students.
He has demonstrated that the following general reaction occurs.

$$[Co(NH_3)_5X]^{2+} + Cr^{2+}_{(aq)} + 5H^+ =$$

$$[Cr(H_2O)_5X]^{2+} + Co^{2+}_{(aq)} + 5NH_4^+$$

[X = F^-, Cl^-, Br^-, I^-, SO_4^{2-}, NCS^-, N_3^-, PO_4^{3-}, $P_2O_7^{4-}$, $CH_3CO_2^-$, $C_3H_7CO_2^-$, succinate, oxalate, malonate]

The Co^{III} complex is not labile, while the Cr^{II} aqua ion is. In the products, the $[Cr(H_2O)_5X]^{2+}$ ion is not labile, whereas the Co^{II} aqua ion is.

It is found that the transfer of X from $[Co(NH_3)_5X]^{2+}$ to $[Cr(H_2O)_5X]^{2+}$ is quantitative. The most reasonable explation —

$$[Cr^{II}(H_2O)_6]^{2+} + [Co^{III}(NH_3)_5Cl]^{2+} \longrightarrow [(H_2O)_5Cr^{II}ClCo^{III}(NH_3)_5]$$

$$\uparrow\downarrow$$

$$[Cr(H_2O)_5Cl]^{2+} + [Co(NH_3)_5H_2O]^{2+} \rightleftharpoons [(H_2O)_5Cr^{III}ClCo^{II}(NH_3)_5]$$

$$\downarrow H^+, H_2O$$

$$[Co(H_2O)_6]^{2+} + 5NH_4^+$$